

MATH 180 Strategies of Problem Solving

W.R. Casper

Department of Mathematics
California State University Fullerton

September 13, 2026

Outline

- 1 SoPS Lecture 5
 - Battleship
 - Probability and Expectation Values
 - Later game
 - Wrapping up

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1 SoPS Lecture 5

- **Battleship**
- Probability and Expectation Values
- Later game
- Wrapping up

Chomp Review

Last time

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- we explored 3×3 Chomp

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- we learned **proof by cases**

Chomp Review

Last time

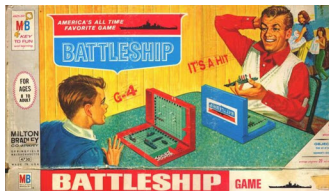
- we explored 3×3 Chomp
- we learned **proof by cases**
- we learned the **Strong Law of Small Numbers** and saw examples

Battleship Introduction



Activity (30 minutes)

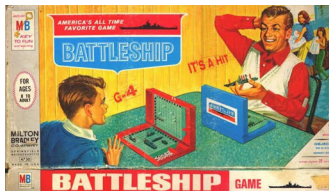
Battleship Introduction



Activity (30 minutes)

- review the rules of Battleship

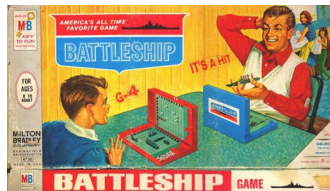
Battleship Introduction



Activity (30 minutes)

- review the rules of Battleship
- split into pairs and play some games!

Battleship Introduction



Activity (30 minutes)

- review the rules of Battleship
- split into pairs and play some games!
- can you observe any useful strategies?

Battleship Discussion

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Question 1

How does the game Battleship differ from other games we have played in this class. How is it similar?

Battleship Discussion

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How does the game Battleship differ from other games we have played in this class. How is it similar?

Question 2

What mathematical tools could be useful in exploring the game Battleship?

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Probability Intro

Vocabulary

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- the set Ω of all possible outcomes is the **sample space**
- subsets $E \subseteq \Omega$ are called **events**
- the **probability** of an event E is

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|}$$

Probability Example

Rolling a Die

Experiment: roll a six-sided die

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- the sample space is:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

- the event of “getting a 3” is $\{3\}$

$$\mathbb{P}(\{3\}) = \frac{|\{3\}|}{|\Omega|} = \frac{1}{6}.$$

Probability Example

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Experiment: roll a six-sided die

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- the event of “getting a 3” is $\{3\}$

$$\mathbb{P}(\{3\}) = \frac{|\{3\}|}{|\Omega|} = \frac{1}{6}.$$

- the event of “getting an even number” is $\{2, 4, 6\}$

$$\mathbb{P}(\{2, 4, 6\}) = \frac{|\{2, 4, 6\}|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}.$$

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Drawing a Card

Experiment: draw a card from a standard deck

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Probability Example

Drawing a Card

Experiment: draw a card from a standard deck

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- what is the event "getting a face card in hearts"?

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$$E = \{(J, \heartsuit), (Q, \heartsuit), (K, \heartsuit)\}$$

- what is the probability of this event?

$$\mathbb{P}(E) = \frac{|E|}{|\Omega|} = \frac{3}{52}.$$

Probability Meaning

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- perform the same experiment a thousand times

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- perform the same experiment a thousand times
- record how many times the outcome is in E
- fraction of times the outcome is in E is approx. $\mathbb{P}(E)$
- more experiments $\rightarrow$ closer to $\mathbb{P}(E)$

Expectations in Battleship

Suppose we play mini-Battleship on a 5×5 board, with a single 3×1 .

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Activity (2 minutes)

Draw a picture of a 5×5 board and secretly place a single 3×1 ship on the board.

Submit your secret board:



Probability in Battleship

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Question 1

How many things are in the sample space?

Probability in Battleship

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Question 1

How many things are in the sample space?

Question 2

What is the probability of a ship being in each spot?

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Suppose we play mini-Battleship on a 5×5 board, with a single 3×1 submarine which is placed randomly.

Expectations in Battleship

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Question 3

How many things are in the sample space?

Expectations in Battleship

Suppose we play mini-Battleship on a 5×5 board, with a single 3×1 submarine which is placed randomly.

Question 3

How many things are in the sample space?

Question 4

What is the probability of a ship being in each spot?

Are humans random?

Activity (5 minutes)

Compare the heat map for the 5×5 with a single 3×1 placed with the heat map obtained from aggregating the data from the class. Discuss similarities or differences with your neighbors.

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Expectations in Battleship

Suppose we take our first shot at the center and we miss.

Expectations in Battleship

Suppose we take our first shot at the center and we miss.

Question 5

How do the probabilities change?

Expectations in Battleship

Suppose we take our second shot at the space one above and to the right of the center and we miss again.

Expectations in Battleship

Suppose we take our second shot at the space one above and to the right of the center and we miss again.

Question 5

How do the probabilities change now?

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Discussion

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In Battleship, we no longer have **perfect information**. There are things one player knows that another does not. This makes tools like probability

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What does probability mean in your own words?

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Question 1

What does probability mean in your own words?

Question 2

How can we use probability to play Battleship better?

Question of the Day

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Puzzle

Suppose we play mini-Battleship on a 5×5 board. How many ways are there to place a 2×1 Destroyer and a 3×1 submarine on the board **simultaneously**.

Final thoughts

Reflection

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- Can you think of any other games where probability may be an important tool? Describe one.

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Reflection

- Can you think of any other games where probability may be an important tool? Describe one.
- What strategies would you suggest for placing the ships in Battleship? What about firing the shots?

Submit your responses by **Tonight!**